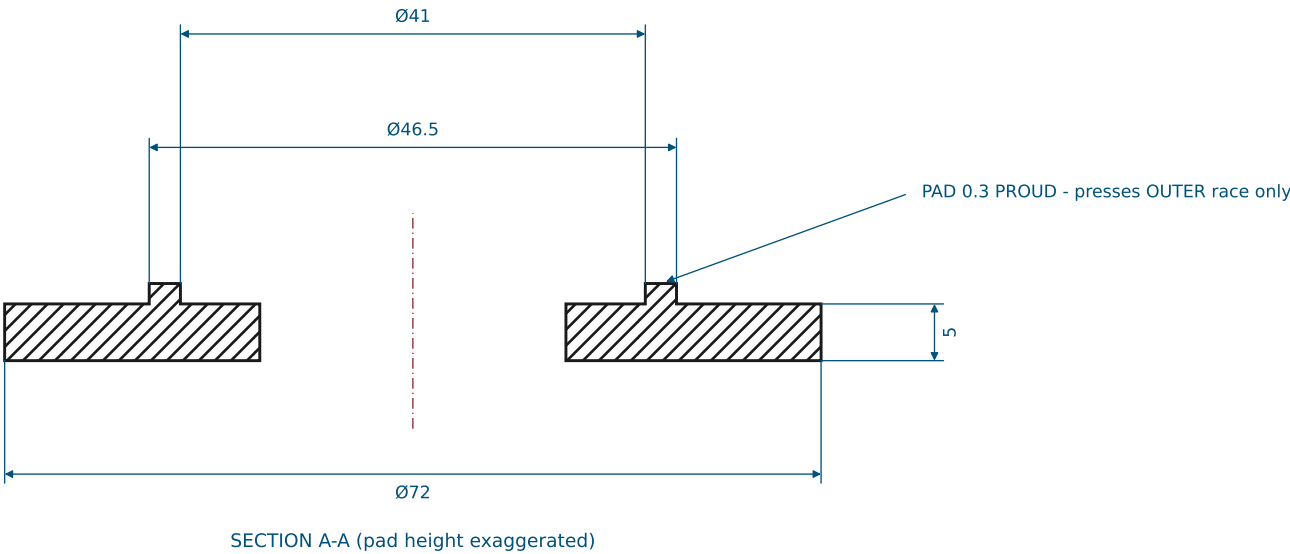
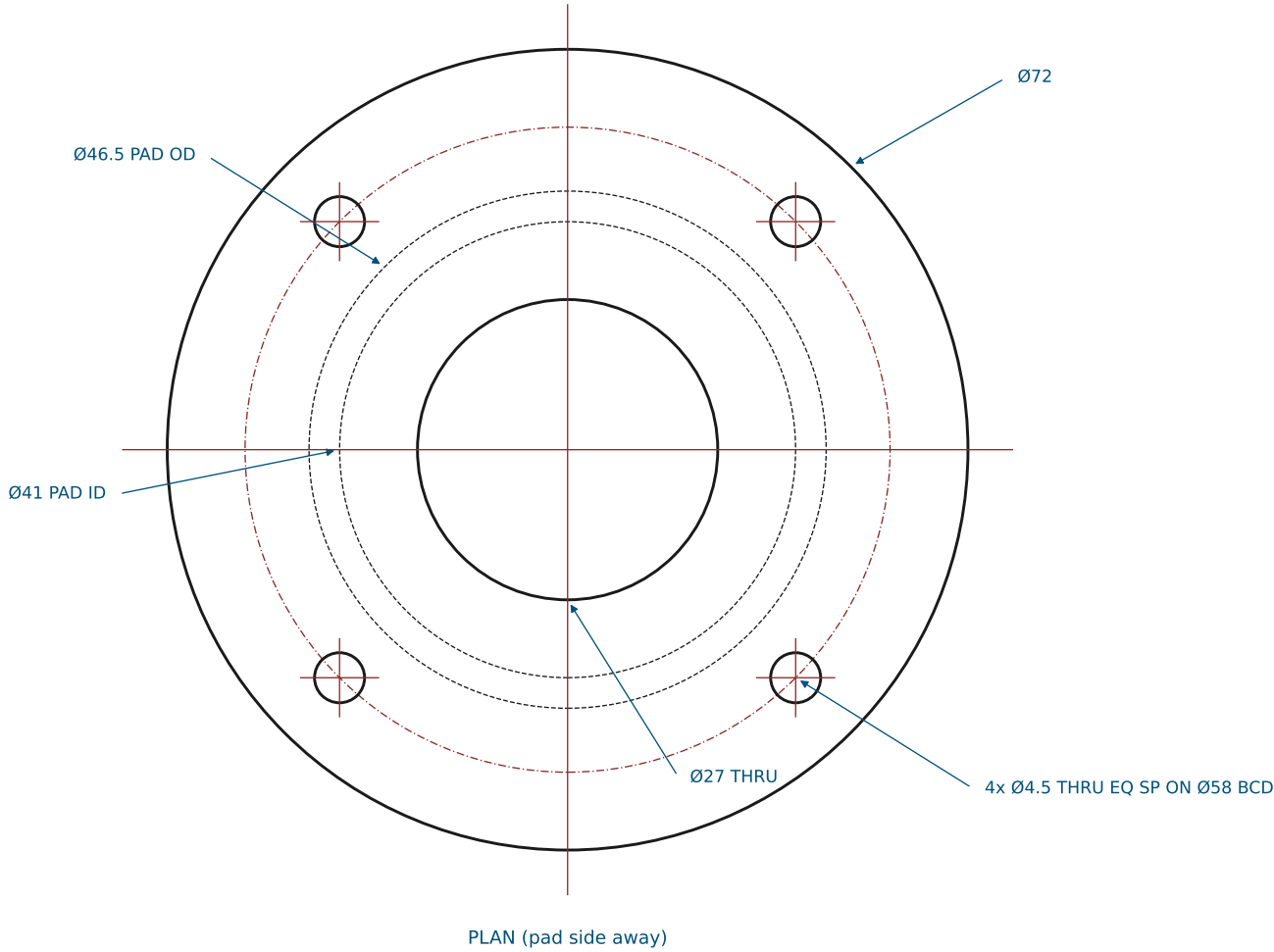


SHEET 1: BEARING RETAINER CAP (BRG-CAP)

GENERAL TOLERANCES UNLESS NOTED: X ±0.5 X.X ±0.2 HOLES +0.2/-0 ANGLES ±0.5° BREAK ALL EDGES 0.3  
DUAL MATERIAL CALLOUTS: select ONE per order — printed parts take brass heat-set inserts at every thread callout; 6061 parts are tapped.

- 1. One cap above AND one below each 7005 bearing bore; the pair bolts together THROUGH the plate with 4x M4x30 SHCS + nyloc (plate holes 4.5 on the same BCD; grip 22 = 5+12+5 at BOTH plates).
- 2. Both plates are 12 THK = the 7005 race width, so the race sits FLUSH with the plate faces and the cap disc seats FLAT on the plate.
- 3. The 0.3-proud pad (41-46.5) lands on the bearing OUTER race only - torquing the bolts gives a small defined clamp crush on the race. No spigot: the cap must NOT stand off the plate.
- 4. Build one cap pad-UP and one cap pad-DOWN per bore (same part, flipped).
- 5. OD is 72 so the bolt holes keep a full 1.5d rim (edge distance 4.75).

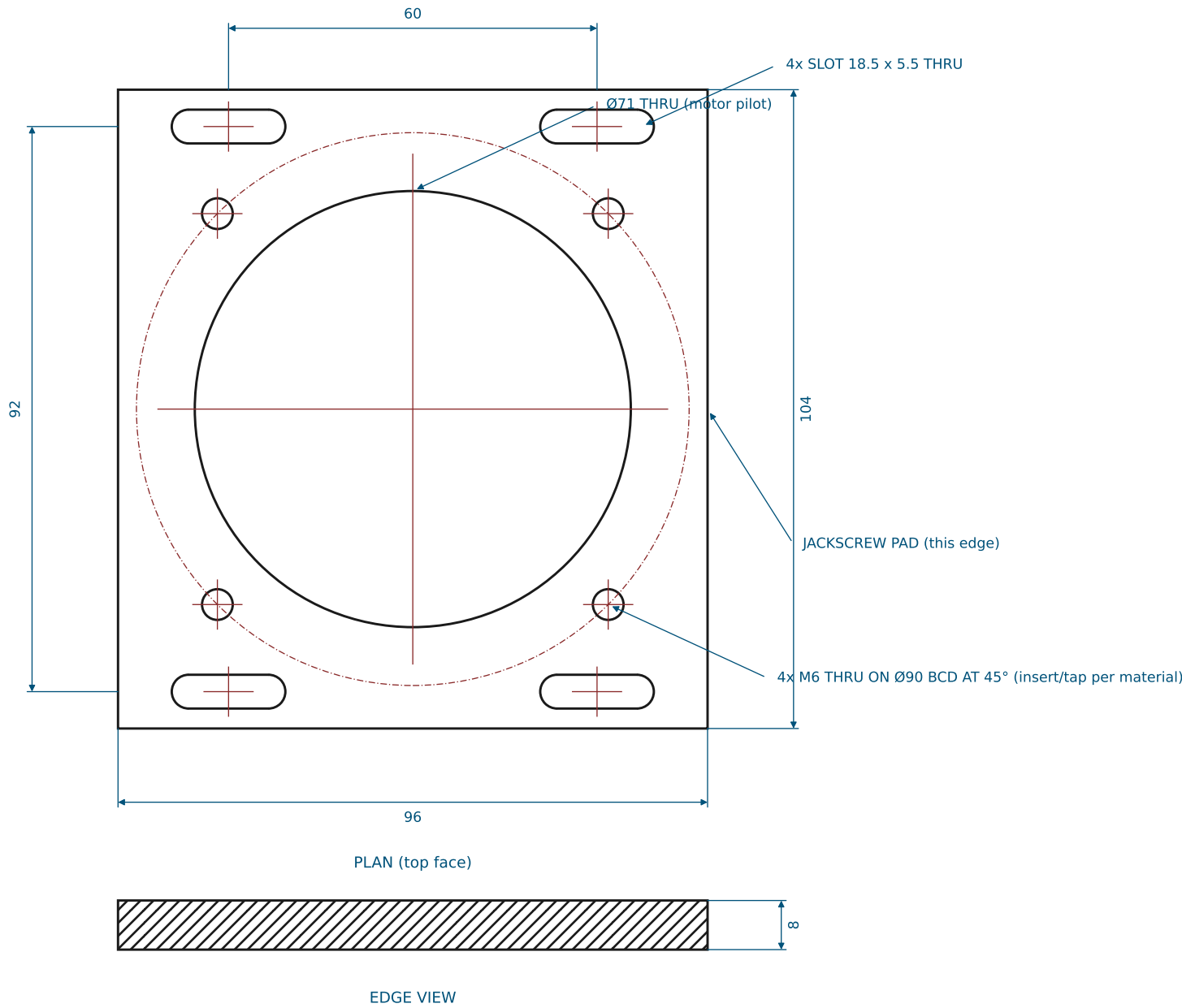


| 5-BAR BUNG-COVER ROBOT — BASE HARDWARE |                         |
|----------------------------------------|-------------------------|
| TITLE: BEARING RETAINER CAP            | DWG NO: BCR-01 REV A    |
| MATERIAL: PA12 MJF / 6061-T6           | QTY: 8                  |
| SCALE 1.5:1 UNITS mm THIRD ANGLE       | SHEET 1 OF 6 2026-07-19 |

SHEET 2: SLIDING MOTOR CARRIAGE (CARRIAGE)

GENERAL TOLERANCES UNLESS NOTED: X ±0.5 X.X ±0.2 HOLES +0.2/-0 ANGLES ±0.5° BREAK ALL EDGES 0.3  
DUAL MATERIAL CALLOUTS: select ONE per order — printed parts take brass heat-set inserts at every thread callout; 6061 parts are tapped.

- 1. Motor (A6M80, 80 sq flange) mounts from BELOW: 4x M6x18 SHCS up through the motor flange into the four M6 positions. PRINTED part: M6 brass heat-set inserts. 6061: tap M6.
- 2. The four slots take M5x16 SHCS + washer down into the cradle frame bosses directly beneath; slot length gives ±6 mm of tension travel.
- 3. The +X edge (toward the shoulder) is the jackscrew pad — the M6 jackscrew tip bears mid-edge.
- 4. Same part both sides (symmetric).

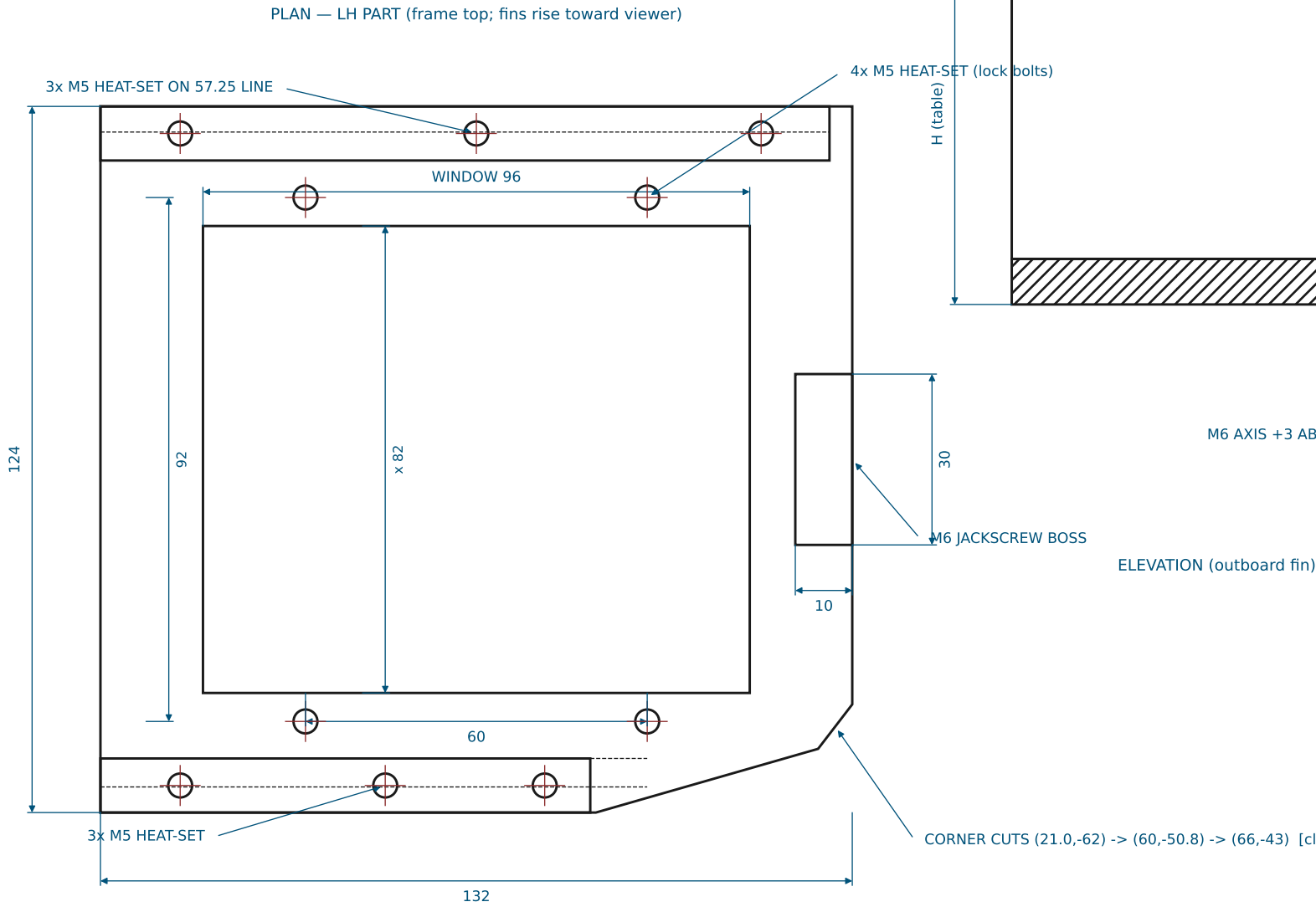


| 5-BAR BUNG-COVER ROBOT — BASE HARDWARE |                         |
|----------------------------------------|-------------------------|
| TITLE: SLIDING MOTOR CARRIAGE          | DWG NO: BCR-02 REV A    |
| MATERIAL: PA12 MJF / 6061-T6           | QTY: 2                  |
| SCALE 1:1 UNITS mm THIRD ANGLE         | SHEET 2 OF 6 2026-07-19 |

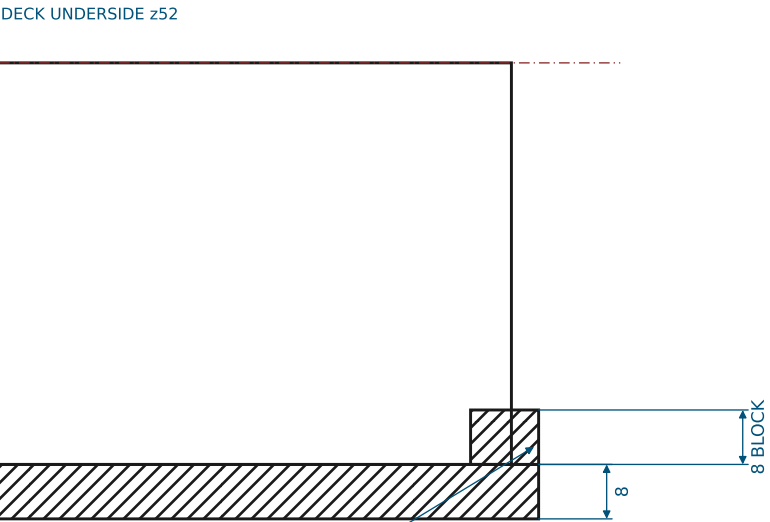
SHEET 3: SLIDER CRADLE (LH SHOWN, RH = MIRROR) (CRADLE-L / CRADLE-R)

GENERAL TOLERANCES UNLESS NOTED: X ±0.5 X.X ±0.2 HOLES +0.2/-0 ANGLES ±0.5° BREAK ALL EDGES 0.3  
DUAL MATERIAL CALLOUTS: select ONE per order — printed parts take brass heat-set inserts at every thread callout; 6061 parts are tapped.

- ONE printed part per motor = window frame + two shear fins + jack block. The RH part is the exact MIRROR of the LH drawn here; fin/wall heights differ per the table (belt planes stagger).
- Fin top flanges (10 thick - full-depth M5 heat-set) bolt UP into the deck underside: 3x M5 per fin, M5x20 SHCS from the deck top (deck holes 5.2 THRU; ~8 mm engagement).
- The four M5 positions in the frame border take the carriage lock bolts — fit heat-set inserts.
- M6 heat-set (or tapped) jackscrew boss in the block; jam nut on the screw against the block face.
- The chamfered corner keeps the two cradles clear of the machine centreline — do not 'square it up'.
- FIN WALLS: inner faces at b = ±52.5, 5 thick, rising to the deck underside. Outboard wall spans a = -66..+62, inboard a = -66..+30. The inner faces guide the carriage edges (0.5 nominal).
- Install the belt around both pulleys BEFORE bolting the deck down (the fins close the corridor).



DATUM: origin at frame centre. +a = toward the shoulder (jack-block end), +b = toward the OUTBOARD fin. All holes in the HOLE TABLE.



|          | FRAME TOP z | WALL H (frame btm→deck) | MIRROR   |
|----------|-------------|-------------------------|----------|
| CRADLE-L | -7          | 67                      | as drawn |
| CRADLE-R | 13          | 47                      | mirrored |

HOLE TABLE (a, b) — LH part; RH = same with b negated

| ID | a      | b      | SPEC                       | ID | a      | b      | SPEC                       |
|----|--------|--------|----------------------------|----|--------|--------|----------------------------|
| H1 | -52.00 | 57.25  | M5 heat-set, fin flange    | L2 | 30.00  | -46.00 | M5 heat-set, carriage lock |
| H2 | 0.00   | 57.25  | M5 heat-set, fin flange    | L3 | -30.00 | 46.00  | M5 heat-set, carriage lock |
| H3 | 50.00  | 57.25  | M5 heat-set, fin flange    | L4 | -30.00 | -46.00 | M5 heat-set, carriage lock |
| H4 | -52.00 | -57.25 | M5 heat-set, fin flange    | J1 | 61.00  | 0.00   | M6 heat-set, jackscrew     |
| H5 | -16.00 | -57.25 | M5 heat-set, fin flange    |    |        |        |                            |
| H6 | 12.00  | -57.25 | M5 heat-set, fin flange    |    |        |        |                            |
| L1 | 30.00  | 46.00  | M5 heat-set, carriage lock |    |        |        |                            |

5-BAR BUNG-COVER ROBOT — BASE HARDWARE

TITLE: SLIDER CRADLE (LH SHOWN, RH = MIRROR)

DWG NO: BCR-03 REV A

MATERIAL: PA12 MJF / PETG-CF (print flat, frame down)

QTY: 1 + 1 (mirror)

SCALE 1:2 / 1:1 UNITS mm THIRD ANGLE

SHEET 3 OF 6 2026-07-19

SHEET 4: BOTTOM DECK PLATE (DECK)

GENERAL TOLERANCES UNLESS NOTED: X ±0.5 X.X ±0.2 HOLES +0.2/-0 ANGLES ±0.5° BREAK ALL EDGES 0.3  
DUAL MATERIAL CALLOUTS: select ONE per order — printed parts take brass heat-set inserts at every thread callout; 6061 parts are tapped.

- 1. Machine per the HOLE TABLE (coordinates from the plate origin = mid-span between the two shoulder bores; +Y toward the arms). All holes THRU.
- 2. The two 47.0 bores carry the 7005 bearings (transition fit: bore 47.0 +0.025/0).
- 3. Corner radii R8. Outline vertices in the VERTEX TABLE.
- 4. Plate must stay within a 300 x 300 print bed (current 294 x 240).

HOLE TABLE

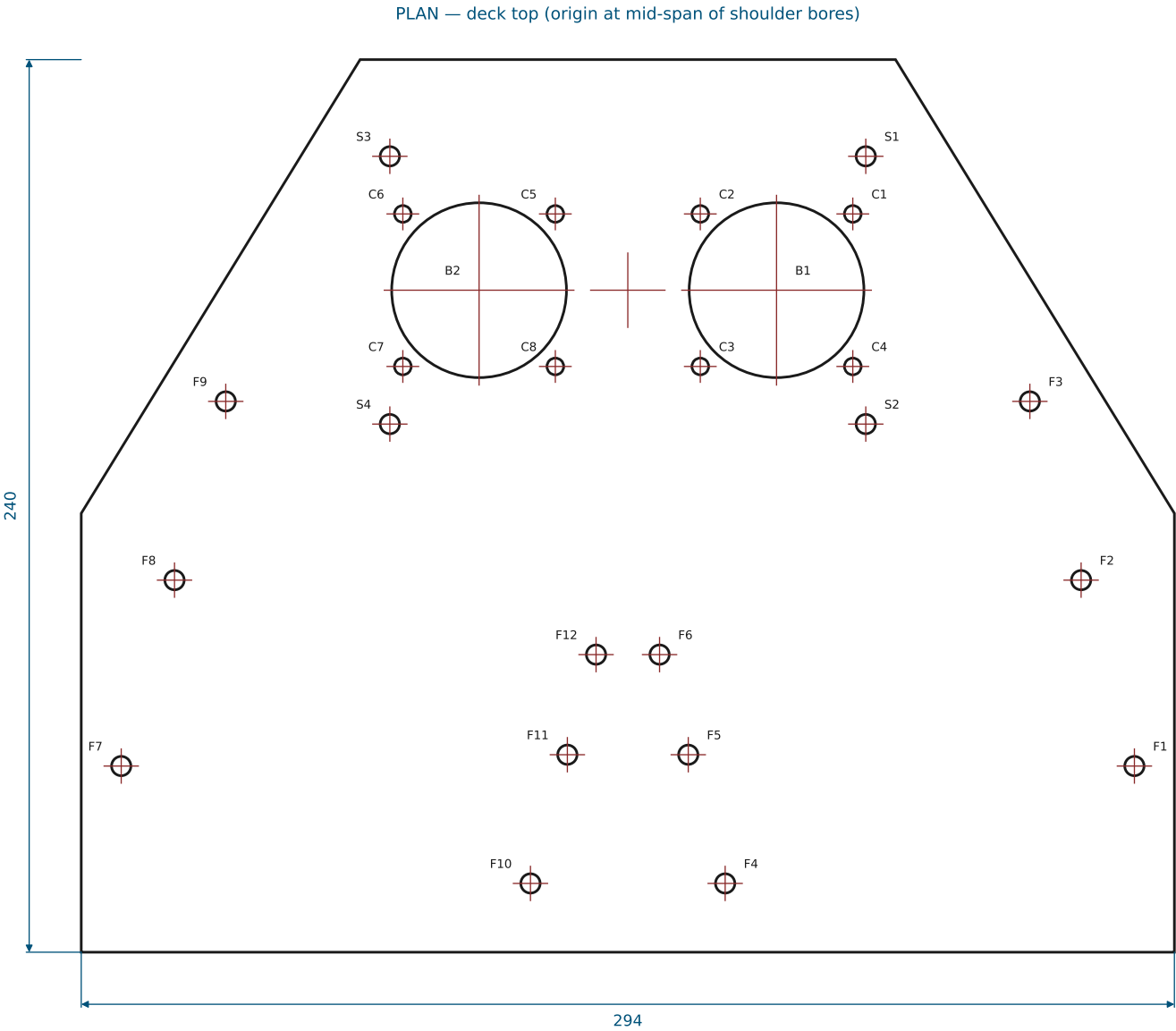
| ID  | X       | Y       | Ø    | SPEC                    |
|-----|---------|---------|------|-------------------------|
| B1  | 40.00   | 0.00    | 47.0 | 7005 bearing bore, thru |
| B2  | -40.00  | 0.00    | 47.0 | 7005 bearing bore, thru |
| C1  | 60.51   | 20.51   | 4.5  | M4 cap bolt, thru       |
| C2  | 19.49   | 20.51   | 4.5  | M4 cap bolt, thru       |
| C3  | 19.49   | -20.51  | 4.5  | M4 cap bolt, thru       |
| C4  | 60.51   | -20.51  | 4.5  | M4 cap bolt, thru       |
| C5  | -19.49  | 20.51   | 4.5  | M4 cap bolt, thru       |
| C6  | -60.51  | 20.51   | 4.5  | M4 cap bolt, thru       |
| C7  | -60.51  | -20.51  | 4.5  | M4 cap bolt, thru       |
| C8  | -19.49  | -20.51  | 4.5  | M4 cap bolt, thru       |
| S1  | 64.00   | 36.00   | 5.2  | M5 standoff bolt, thru  |
| S2  | 64.00   | -36.00  | 5.2  | M5 standoff bolt, thru  |
| S3  | -64.00  | 36.00   | 5.2  | M5 standoff bolt, thru  |
| S4  | -64.00  | -36.00  | 5.2  | M5 standoff bolt, thru  |
| F1  | 136.25  | -127.95 | 5.2  | M5 fin bolt, thru       |
| F2  | 121.91  | -77.96  | 5.2  | M5 fin bolt, thru       |
| F3  | 108.13  | -29.90  | 5.2  | M5 fin bolt, thru       |
| F4  | 26.18   | -159.51 | 5.2  | M5 fin bolt, thru       |
| F5  | 16.26   | -124.90 | 5.2  | M5 fin bolt, thru       |
| F6  | 8.54    | -97.99  | 5.2  | M5 fin bolt, thru       |
| F7  | -136.25 | -127.95 | 5.2  | M5 fin bolt, thru       |
| F8  | -121.91 | -77.96  | 5.2  | M5 fin bolt, thru       |
| F9  | -108.13 | -29.90  | 5.2  | M5 fin bolt, thru       |
| F10 | -26.18  | -159.51 | 5.2  | M5 fin bolt, thru       |
| F11 | -16.26  | -124.90 | 5.2  | M5 fin bolt, thru       |
| F12 | -8.54   | -97.99  | 5.2  | M5 fin bolt, thru       |

VERTEX TABLE (outline, R8 corners)

|               |                |               |
|---------------|----------------|---------------|
| V1 (-72,62)   | V2 (72,62)     | V3 (147,-60)  |
| V4 (147,-178) | V5 (-147,-178) | V6 (-147,-60) |

5-BAR BUNG-COVER ROBOT — BASE HARDWARE

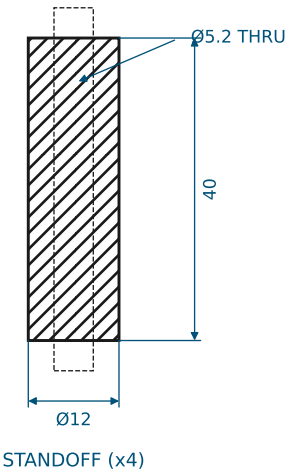
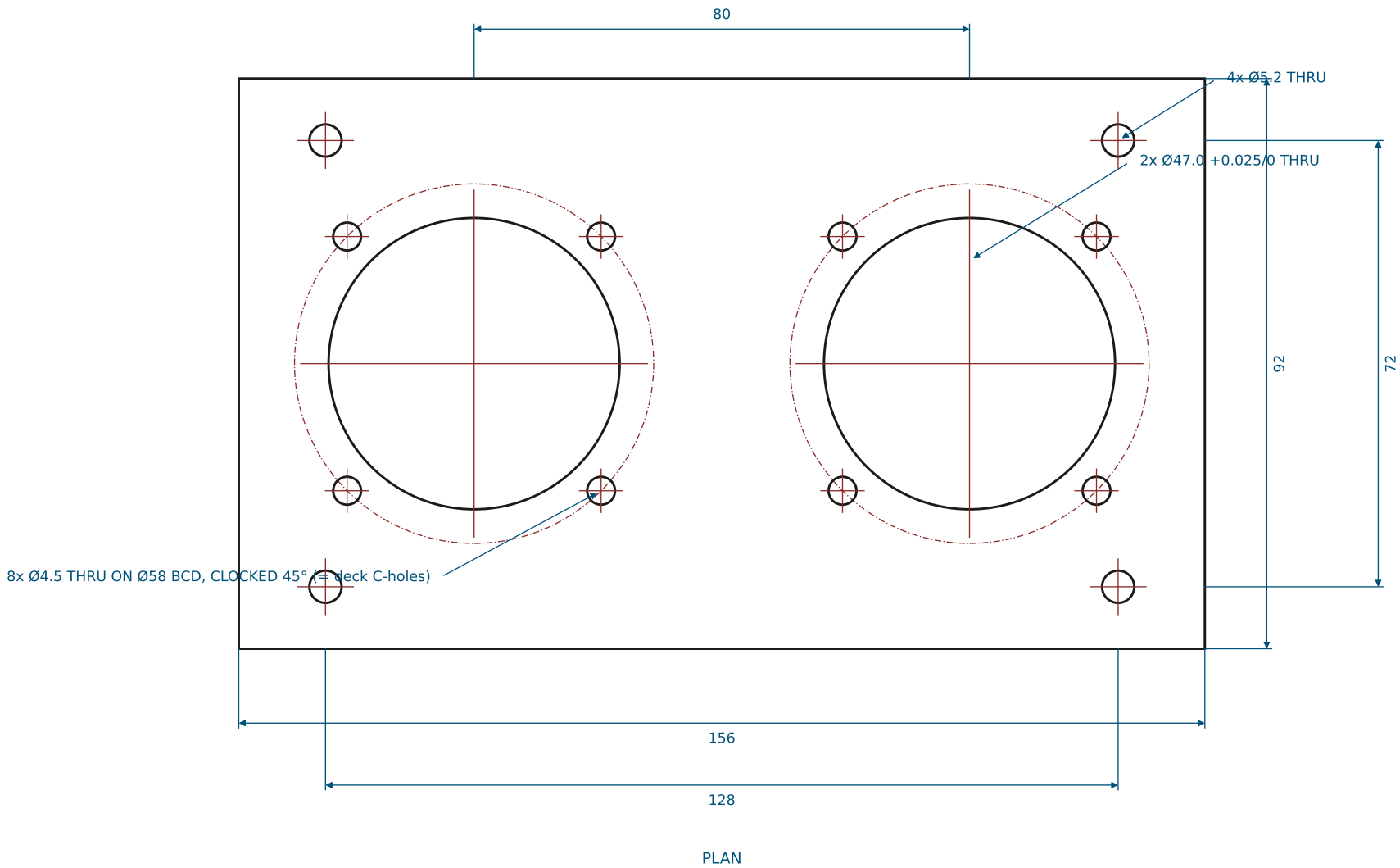
|                                       |                         |
|---------------------------------------|-------------------------|
| TITLE: BOTTOM DECK PLATE              | DWG NO: BCR-04 REV A    |
| MATERIAL: PA12 MJF / 6061-T6 (12 THK) | QTY: 1                  |
| SCALE 1:2 UNITS mm THIRD ANGLE        | SHEET 4 OF 6 2026-07-19 |



SHEET 5: TOP PLATE + STANDOFFS (TOP-PLATE)

GENERAL TOLERANCES UNLESS NOTED: X ±0.5 X.X ±0.2 HOLES +0.2/-0 ANGLES ±0.5° BREAK ALL EDGES 0.3  
DUAL MATERIAL CALLOUTS: select ONE per order — printed parts take brass heat-set inserts at every thread callout; 6061 parts are tapped.

- 1. Same bearing bore + cap-bolt pattern as the deck (47.0 bores, 4x 4.5 on Ø58 BCD per bore). 12 THK = the 7005 race width: the race sits flush with both faces (flat caps seat on the plate).
- 2. Standoff bolt holes moved OUT to (±64, ±36) so the Ø72 caps clear the standoff bosses (they collided with the old Ø62 caps at ±60,±32).
- 3. 4x standoffs Ø12 x 40, Ø5.2 clearance bore THRU. Each stack is THROUGH-BOLTED: one M5 x 75 SHCS from the plate top, nyloc + washer under the deck (no threads in the standoff; grip 64).

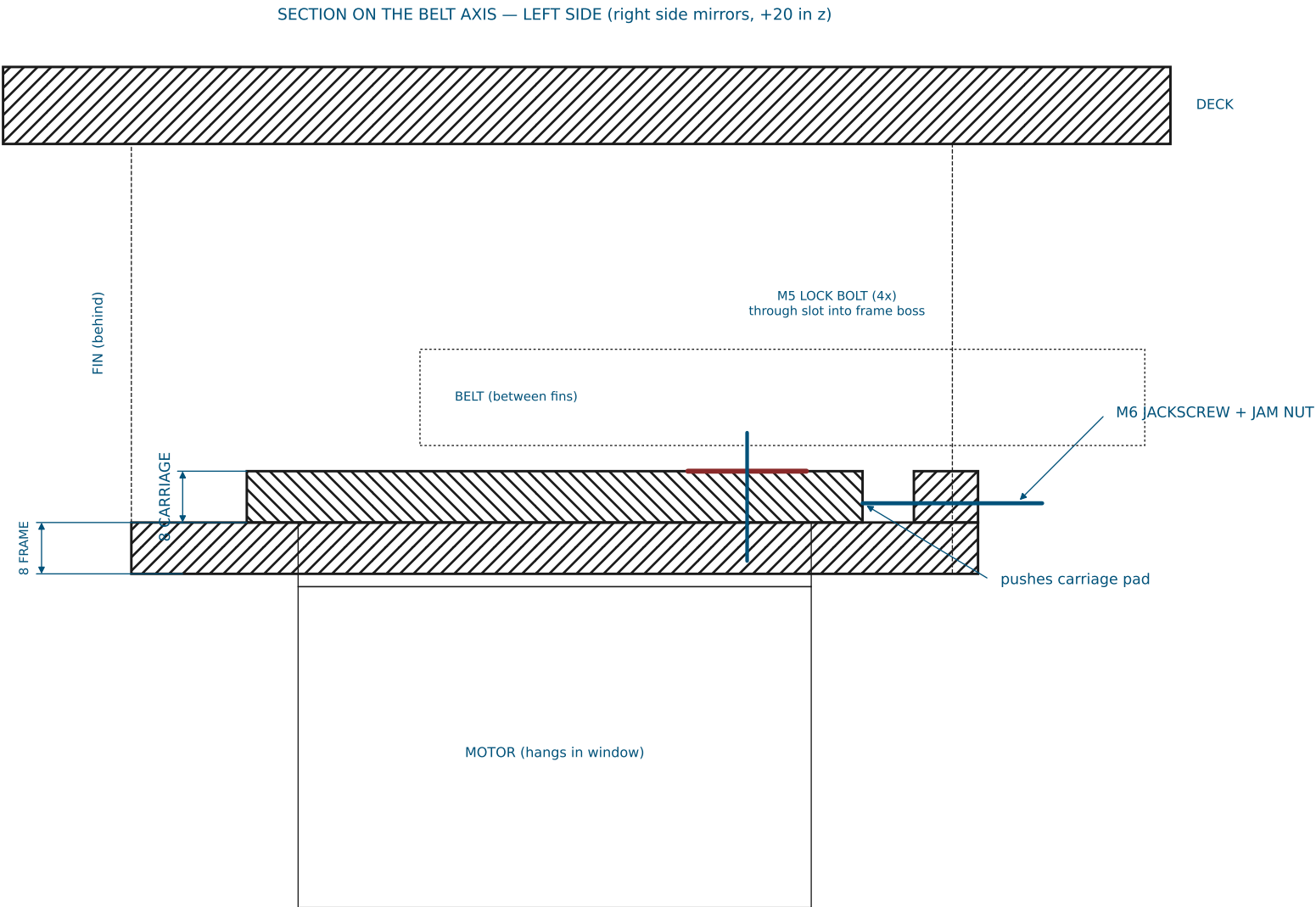


| 5-BAR BUNG-COVER ROBOT — BASE HARDWARE                  |                         |
|---------------------------------------------------------|-------------------------|
| TITLE: TOP PLATE + STANDOFFS                            | DWG NO: BCR-05 REV A    |
| MATERIAL: PA12 MJF / 6061-T6 (12 THK = 7005 race width) | QTY: 1 (+4 standoffs)   |
| SCALE 1:1 UNITS mm THIRD ANGLE                          | SHEET 5 OF 6 2026-07-19 |

SHEET 6: TENSIONER ASSEMBLY + HARDWARE (ASSY-TENSIONER)

GENERAL TOLERANCES UNLESS NOTED: X ±0.5 X.X ±0.2 HOLES +0.2/-0 ANGLES ±0.5° BREAK ALL EDGES 0.3  
DUAL MATERIAL CALLOUTS: select ONE per order — printed parts take brass heat-set inserts at every thread callout; 6061 parts are tapped.

1. TENSION PROCEDURE: loosen the 4x M5 lock bolts 1/2 turn -> turn the M6 jackscrew CW to push the carriage away from the shoulder (belt tightens) -> torque the lock bolts (M5: 4 N·m in inserts) -> set the M6 jam nut against the block.
2. Travel available ±6 mm. Belt: 450-5M-15 (24T -> 72T, C = 97.5 nominal).
3. ASSEMBLY ORDER: pulleys + belt on motor/shaft FIRST, then lower the deck onto fins + standoffs and bolt down (the fin walls close the belt corridor once the deck is on).
4. The belt passes BETWEEN the two fins, above the frame and below the deck — nothing crosses it.



HARDWARE (per machine)

- 16x M4 x 30 SHCS + nyloc + washers  
bearing cap pairs, 4 per bore stack (grip 22 both plates)
- 8x M5 x 16 SHCS + washer  
carriage lock bolts (4 per side)
- 12x M5 x 20 SHCS  
fin flanges -> deck underside (10-deep flange, full heat-set)
- 4x M5 x 75 SHCS + nyloc  
top plate -> standoff -> deck through-bolt (one per standoff, grip 64)
- 8x M6 x 18 SHCS  
motor flange -> carriage from below (full 8 mm engagement)
- 2x M6 x 40 jackscrew + jam nut  
tensioner
- 2x M25x1.5 shaft locknut (KM5)  
bearing preload
- M5/M6 brass heat-set inserts  
all printed bosses

| 5-BAR BUNG-COVER ROBOT — BASE HARDWARE |                         |
|----------------------------------------|-------------------------|
| TITLE: TENSIONER ASSEMBLY + HARDWARE   | DWG NO: BCR-06 REV A    |
| MATERIAL: -                            | QTY: 2 (mirrored)       |
| SCALE 1:1 section UNITS mm THIRD ANGLE | SHEET 6 OF 6 2026-07-19 |